

Generation Interconnection System Impact Study Report

For Queue Project Q21-1871

Cedar Hill 100 kV

June 20, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1871. The Interconnection Customer proposes a 174.7 MW solar photovoltaic facility interconnecting at the Cedar Hill 100 kV substation in the Ohio area. The request was received on August 28, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 489 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,420,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: FOX to CEDAR HILL 100kV
Contingency: IMPERIAL to EUFAULA 500kV Line
Loading: 161.7% (Rating: 163 MVA)
DFAX: 58.1%
Pre-project loading: 99.4%
2. Facility: POUST to FOX 100kV line
Contingency: IMPERIAL to EUFAULA 500kV
Loading: 159.0% (Rating: 132 MVA)
DFAX: 32.1%
Pre-project loading: 116.5%
3. Facility: W UNION to WELLINGTON 1 100kV line
Contingency: IMPERIAL to EUFAULA 500kV
Loading: 147.4% (Rating: 148 MVA)
DFAX: 10.2%
Pre-project loading: 135.4%
4. Facility: WELLINGTON 1 to FOWLES 100kV
Contingency: IMPERIAL to EUFAULA SUBSTATION 500kV L
Loading: 147.3% (Rating: 173 MVA)
DFAX: 10.2%
Pre-project loading: 137.1%
5. Facility: CLEVELAND THERMAL to MARCOS 100kV line
Contingency: MARCOS SUBSTATION to SCRANTON 100kV line
Loading: 144.8% (Rating: 165 MVA)
DFAX: 7.1%
Pre-project loading: 137.2%
6. Facility: SCRANTON SUBSTATION to MARCOS 100kV
Contingency: CLEVELAND THERMAL SUBSTATION to MARCOS 100kV
Loading: 137.1% (Rating: 152 MVA)

DFAX: 6.0%

Pre-project loading: 130.2%

7. Facility: IX CTR to FOWLES 100kV

Contingency: GETTY to COSUMNES 100kV

Loading: 133.6% (Rating: 132 MVA)

DFAX: 11.8%

Pre-project loading: 118.0%

8. Facility: WEST 41ST STREET to LORING POWER PLANT 100kV

Contingency: GALEVILLE SUBSTATION to MAROS 100kV

Loading: 119.9% (Rating: 112 MVA)

DFAX: 7.0%

Pre-project loading: 108.9%

9. Facility: HILLS to LINCOLN BEACH 100kV

Contingency: IMPERIAL to CHAMBERLIN 230kV

Loading: 111.2% (Rating: 179 MVA)

DFAX: 7.6%

Pre-project loading: 103.8%

10. Facility: HILLS to NORDONIA 100kV

Contingency: IMPERIAL to CHAMBERLIN 230kV

Loading: 110.9% (Rating: 157 MVA)

DFAX: 6.5%

Pre-project loading: 103.6%

11. Facility: GALEVILLE to MARCOS 100kV

Contingency: EUFAULA to IMPERIAL SUBSTATION 500kV

Loading: 110.2% (Rating: 297 MVA)

DFAX: 6.5%

Pre-project loading: 106.3%

12. Facility: BACA to POUST 100kV L

Contingency: WELLINGTON 1 to FOWES 100kV

Loading: 104.7% (Rating: 136 MVA)

DFAX: 10.6%

Pre-project loading: 91.1%

13. Facility: WEST UNION to LORING POWER PLANT SUBSTATION 100kV

Contingency: IMPERIAL to EUFAULA SUBSTATION 500kV

Loading: 102.7% (Rating: 197 MVA)

DFAX: 5.8%

Pre-project loading: 97.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FOX to CEDAR HILL 100kV	Rebuild line with higher-capacity conductor	\$2,005,000	92% (\$1,838,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
POUST to FOX 100kV line	Rebuild line with higher-capacity conductor	\$4,510,000	83% (\$3,736,000)
WEST UNION to WELLINGTON 1 100kV	Rebuild line with higher-capacity conductor	\$2,249,000	63% (\$1,412,000)
WLLINGTON 1 to FOWLES 100kV	Rebuild line with higher-capacity conductor	\$1,893,000	63% (\$1,188,000)
CLEVELAND THERMAL to MARCOS 100kV line	Rebuild line with higher-capacity conductor	\$1,893,000	48% (\$919,000)
SCRANTON to MARCOS 100kV	Rebuild line with higher-capacity conductor	\$2,208,000	78% (\$1,719,000)
IX CTR to FOWLES 100kV	Rebuild line with higher-capacity conductor	\$2,048,000	100% (\$2,048,000)
LORING POWER PLANT to WEST 41ST ST 100kV line	Rebuild line with higher-capacity conductor	\$4,418,000	100% (\$4,418,000)
HILLS to LINCOLN BEACH 100kV	Rebuild line with higher-capacity conductor	\$2,347,000	67% (\$1,581,000)
NORONIA SUBSTATION to HILLS 100kV	Rebuild line with higher-capacity conductor	\$6,247,000	67% (\$4,189,000)
GALEVILLE to MARCOS 100kV Line	Rebuild line with higher-capacity conductor	\$4,894,000	80% (\$3,912,000)
POUST to BACA 100kV	Reconductor line	\$1,147,000	100% (\$1,147,000)
WEST UNION to LORING POWER PLANT SUBSTATION 100kV	Reconductor line	\$649,000	48% (\$313,000)
	Total allocated to project		\$28,420,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.